11.11.2023

Tema 03

Ex. 1:

$b_1 b_2 \dots b_m$ bits

$\mathbb{P}(A_1) = 0$

$\mathbb{P}(A_2) = 0$

$\mathbb{P}(A_3) = 0$

$\mathbb{P}(A_4) = \frac{1}{16} = \frac{1}{2^4}$

$$\mathbb{P}(A_5) = \frac{1}{2}\mathbb{P}(A_5 \mid b_5 = 0) + \frac{1}{2}\mathbb{P}(A_5 \mid b_5 = 1) =$$

$$= \frac{1}{2}\mathbb{P}(A_4) + \frac{1}{2}\left[\frac{1}{2}\mathbb{P}(A_5 \mid b_4 b_5 = 01) + \frac{1}{2}\mathbb{P}(A_5 \mid b_4 b_5 = 11)\right] =$$

$$= \frac{1}{2}\mathbb{P}(A_4) + \frac{1}{4}\mathbb{P}(A_3) + \frac{1}{4}\left[\frac{1}{2}\mathbb{P}(A_5 \mid b_3 b_4 b_5 = 011) + \frac{1}{2}\mathbb{P}(A_5 \mid b_3 b_4 b_5 = 111)\right] =$$

$$= \frac{1}{2}\mathbb{P}(A_4) + \frac{1}{4}\mathbb{P}(A_3) + \frac{1}{8}\mathbb{P}(A_2) +$$

$$+ \frac{1}{8}\left[\frac{1}{2}\underbrace{\mathbb{P}(A_5 \mid b_2 b_3 b_4 b_5 = 1111)}_{=1} + \frac{1}{2}\mathbb{P}(A_5 \mid b_2 b_3 b_4 b_5 = 0111)\right] =$$

$$= \frac{1}{2}\mathbb{P}(A_4) + \frac{1}{4}\mathbb{P}(A_3) + \frac{1}{8}\mathbb{P}(A_2) + \frac{1}{16}\mathbb{P}(A_1) + \frac{1}{16}\cdot 1$$

Analog demonstrăm formula generală:

$$\mathbb{P}(A_m) = \frac{1}{2}\mathbb{P}(A_{m-1}) + \frac{1}{4}\mathbb{P}(A_{m-2}) + \frac{1}{8}\mathbb{P}(A_{m-3}) +$$

$$+ \frac{1}{16}\mathbb{P}(A_{m-4}) + \frac{1}{16}$$